Conditional All-Level Reset Dichotomy

Inclusion-Minimal Native and Adaptive-Lag Interfaces

Prime Dynamics Theory Program

July 2026

Abstract

The reset program now has two finite survivors but no all-level theorem. We turn that boundary into a precise conditional dichotomy. Let U_n be rank-r spectral reset frames, $r \geq 4$, let $M_n = R_n + T_n$, and denote the selected full-memory endpoints by ℓ_n, u_n, a tail-mass upper by τ_n, and the consecutive frame overlap by $C_n = U_{n-1}^* U_n$.

Three eventual interfaces suffice for a uniform native floor:

$$\sigma_{\min}(C_n) \geq a > 0, \qquad q_n := \frac{\tau_n}{\ell_n - \tau_n} \leq q < 1, \qquad \delta_n := \frac{\ell_n - \tau_n}{u_n} \geq \delta > 0.$$

They imply

$$\inf_n \mathcal{N}_n \geq a\sqrt{\delta}(1 - \sqrt{q})^4 > 0.$$

If, additionally, some lag $1 \leq k_n \leq K$ has normalized fourth-cross lower $\sigma_4((I - P_{n-k_n})R_n P_{n-k_n})/\sigma_1 \geq b > 0$, then the transported directional seed obeys $\inf_n \mathcal{D}_n \geq a^K b > 0$.

The native set is inclusion-minimal for this formula; the joint native–directional set is inclusion-minimal after the lag interface is split into bounded lag and positive normalized cross. Explicit witnesses make each omitted floor tend to zero. On the frozen atlas every clause passes. Independent worst constants give a native floor 3.0883×10^{-12}, while the selected-path directional floor is 2.1194×10^{-23}; replacing selected paths by a^8 gives only 6.0283×10^{-48}. These are finite calibrations, not evidence that the eventual interfaces hold. The theorem is a falsifiable roadmap and makes no Stage A, Hilbert–Polya, zero-identification, or RH claim.

1 The two output types

Let $P_n = U_n U_n^*$ and compress the full, recent, and tail memories to the current packet:

$$G_n = U_n^* M_n U_n, \qquad H_n = U_n^* R_n U_n, \qquad D_n = U_n^* T_n U_n, \qquad G_n = H_n + D_n.$$

Assume G_n has spectrum in $[\ell_n, u_n]$ and $0 \preceq D_n \preceq \tau_n I$. The native certificate functional is

$$\mathcal{N}_n = \frac{\sigma_{\min}(C_n)}{\sigma_1(C_n)} \sqrt{\frac{\lambda_{\min}(H_n)}{\lambda_{\max}(G_n)}} \left(1 - \sqrt{\|H_n^{-1/2} D_n H_n^{-1/2}\|}\right)_+^4.$$

It packages transition conditioning, recent spectral spread, and relative tail loss exactly as in the RH-156 support formula.

For a lag k, put $Q_{n,k} = P_{n-k}$ and

$$B_{n,k} = (I - Q_{n,k}) R_n Q_{n,k}, \qquad W_{n,k} = C_{n-k+1} \cdots C_n.$$

The transported directional seed is

$$\mathcal{D}_{n,k} = \sigma_{\min}(W_{n,k}) \frac{\sigma_4(B_{n,k})}{\sigma_1(B_{n,k})}.$$

This definition keeps the native and cross types separate. A later assembly must declare which one it accepts.

2 Conditional all-level theorem

Theorem 1 (Native–directional reset dichotomy). *Suppose that, for every $n \geq N$:*

(O) $\sigma_{\min}(C_n) \geq a > 0$;

(E) $\ell_n > \tau_n$ *and* $q_n = \tau_n/(\ell_n - \tau_n) \leq q < 1$;

(S) $(\ell_n - \tau_n)/u_n \geq \delta > 0$.

Then

$$\mathcal{N}_n \geq \mathcal{N}_* := a\sqrt{\delta}(1 - \sqrt{q})^4 > 0 \qquad (n \geq N).$$

If, in addition,

(L) *there are constants $K < \infty$ and $b > 0$ such that for every $n \geq N$ some $1 \leq k_n \leq K$ satisfies*

$$\frac{\sigma_4(B_{n,k_n})}{\sigma_1(B_{n,k_n})} \geq b,$$

then

$$\mathcal{D}_{n,k_n} \geq \mathcal{D}_* := a^K b > 0.$$

Proof. Since $G_n \succeq \ell_n I$ and $D_n \preceq \tau_n I$, $H_n \succeq (\ell_n - \tau_n)I$. Therefore

$$D_n \preceq \frac{\tau_n}{\ell_n - \tau_n} H_n \preceq q H_n, \qquad \frac{\lambda_{\min}(H_n)}{\lambda_{\max}(G_n)} \geq \delta.$$

Frame overlaps are contractions, so $\sigma_1(C_n) \leq 1$ and $\sigma_{\min}(C_n)/\sigma_1(C_n) \geq a$. Substitution into $\mathcal{N}_n$ gives the native floor.

For the selected lag, least singular values are supermultiplicative: $\sigma_{\min}(W_{n,k_n}) \geq a^{k_n} \geq a^K$. Multiplication by the normalized fourth-cross lower b gives the directional floor. □

Condition (E) is the sharp subunit gate: $q < 1$ is equivalent to $\ell_n > 2\tau_n$. Condition (S) is independent; a weak mode and its tail may decay together so that q_n stays below one while the recent/full normalized base tends to zero.

Corollary 2 (Outwardly checkable lag interface)**.** *Let a nominal lagged cross have first and fourth singular values $\widehat{s}_1, \widehat{s}_4$ and certified operator radius ρ. If*

$$\widehat{s}_4 - \rho \geq \gamma > 0, \qquad \widehat{s}_1 + \rho \leq \Gamma < \infty,$$

then (L) holds with $b = \gamma/\Gamma$. For a Hermitian recent center $\widehat{R}$, matrix radius r_R, and projector radius ϵ, one may take

$$\rho = r_R + \big(\lambda_{\max}(\widehat{R}) - \lambda_{\min}(\widehat{R})\big)\epsilon.$$

Proof. Singular-value perturbation gives the first assertion. The second is the shift-invariant centered projector-action radius of RH-158 [Bhatia, 1997, Stewart and Sun, 1990]. □

3 Inclusion-minimal interfaces

The minimality below is relative to the displayed modular functionals, not a claim that no future construction can bypass them.

Theorem 3 (Omission witnesses)**.** *The interface set $\{\mathrm{O}, \mathrm{E}, \mathrm{S}\}$ is inclusion-minimal for a uniform positive lower on $\mathcal{N}_n$. The set $\{\mathrm{O}, \mathrm{E}, \mathrm{S}, \mathrm{L}\}$ is inclusion-minimal for the joint conclusion $\inf \mathcal{N}_n > 0$ and $\inf \mathcal{D}_{n,k_n} > 0$. Within (L), both bounded lag and a positive normalized fourth-cross lower are necessary for the stated directional formula.*

Proof. Keep all unmentioned interface constants fixed. Without (O), take admissible overlap lowers $a_n \downarrow 0$. Without (E), take relative tail ratios $q_n \uparrow 1$ while the recent/full ratio stays bounded below; the fourth power tail factor tends to zero. Without (S), take recent compressions with one eigenvalue $\varepsilon_n \downarrow 0$ and tail mass proportional to ε_n; then q_n remains fixed but $\delta_n \downarrow 0$.

Without a positive cross ratio, choose R_n block diagonal relative to every candidate packet; native positivity can remain uniform while $\sigma_4(B_{n,k}) = 0$. Finally, if only lags $k_n \to \infty$ carry a fixed cross ratio and $0 < a < 1$, then the guaranteed transport factor $a^{k_n} \to 0$. Each proper omission therefore admits a sequence with the remaining modular interfaces intact and the requested floor vanishing. □

The explicit audit uses indices $1, 2, 4, 8, 16, 32, 64$ for all five witnesses and checks monotone floor collapse. These examples mark exact logical failure modes rather than predicting which one the prime-dynamics sequence will exhibit.

4 Finite calibration

The 120-transition atlas supplies

$$a = 8.98663 \times 10^{-5}, \qquad q = 0.2246959, \qquad \delta = 2.01608 \times 10^{-13}.$$

Using independent worst cases in Theorem 1 gives

$$\mathcal{N}_* = 3.08834 \times 10^{-12}.$$

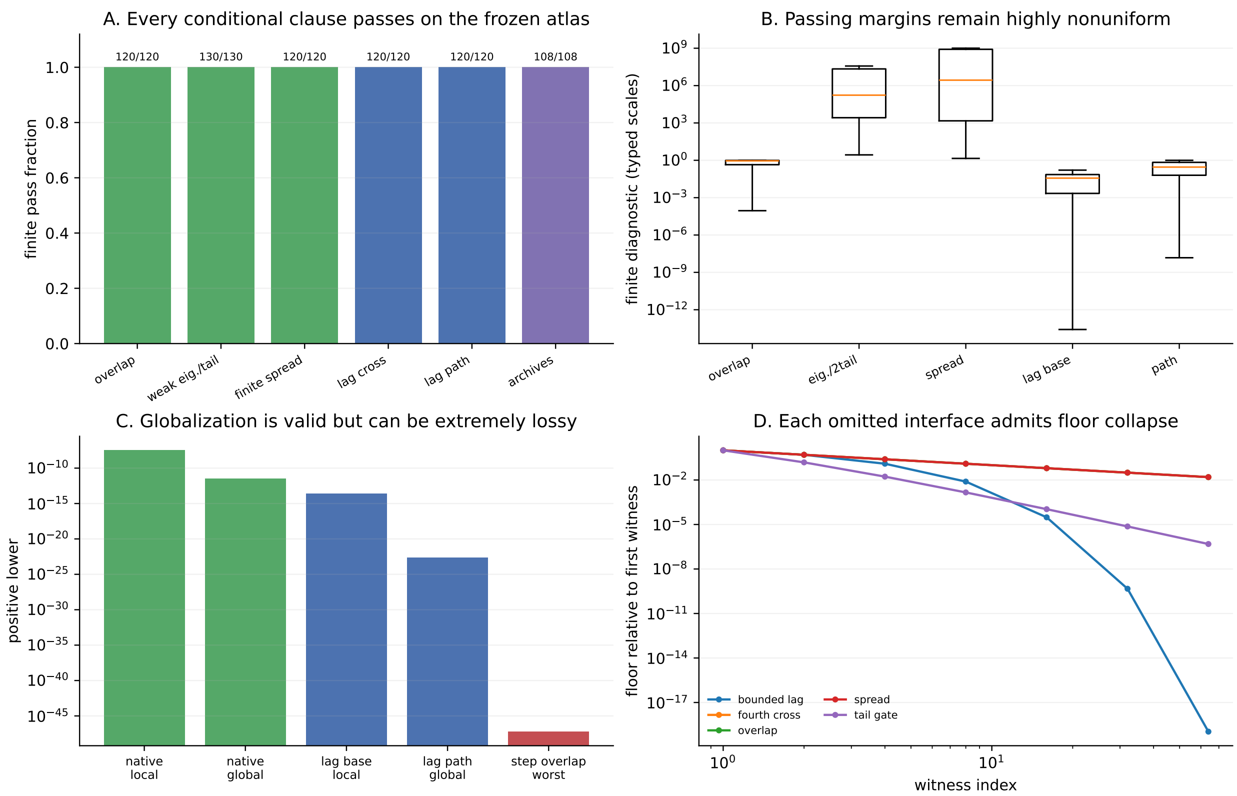

Figure 1: Finite clause audit, nonuniform diagnostics, globalization loss, and the five explicit omission mechanisms.

Taking global endpoint extrema rather than the three interface extrema gives the still more conservative 2.44778×10^{-12}; evaluating each transition locally gives minimum 3.26204×10^{-8}. All are valid finite numbers with different correlation loss.

For the lag route,

$$K = 8, \qquad \gamma = 3.18077 \times 10^{-16}, \qquad \Gamma = 0.224446.$$

The theorem's purely consecutive bound is $a^8\gamma/\Gamma = 6.02832 \times 10^{-48}$. The archived selected paths have common overlap 1.49553×10^{-8}, improving the transported floor to 2.11941×10^{-23}. The minimum local, untransported normalized base is 2.58742×10^{-14}. The forty-order spread between local and modular worst-case bounds is precisely the conditioning problem an all-level proof must address.

5 Falsifiable roadmap and boundary

The next theoretical work is not another large finite count. It is to derive scale laws for O, E, and S, then L if the directional output is required. At each stage the table supplies a clean negative outcome. Only after one route has eventual interfaces should its seed be inserted into a single typed assembly A.

interface	eventual statement	direct falsifier
O	$\inf \sigma_{\min}(C_n) > 0$	running overlap minimum tends to zero
E	$\sup q_n < 1$	weak eigenvalue reaches twice tail mass
S	$\inf \delta_n > 0$	selected recent/full ratio tends to zero
L	bounded lag and $\inf \sigma_4/\sigma_1 > 0$	required lag grows or cross margin closes
A	typed downstream assembly accepts the seed	radius overflow or packet-type mismatch

Table 1: Interfaces and observations that would reject the present route.

The archive audit rechecks all 108 publication hashes from RH-151–RH-159; all match. RH-160 proves the conditional formulas and their omission witnesses and calibrates every hypothesis on the finite atlas. It does not prove that O, E, S, L, or A holds eventually for the prime-dynamics sequence, does not close Stage A, does not construct a Hilbert–Polya operator, does not identify zeta zeros, and does not prove the Riemann Hypothesis.

References

Rajendra Bhatia. *Matrix Analysis*. Springer, 1997.

G. W. Stewart and Ji-Guang Sun. *Matrix Perturbation Theory*. Academic Press, 1990.